Jesse McDonald

jessemcdonald144@gmail.com · 646-573-2663

Cornell University - Ithaca, NY **Graduated May 2026**
Bachelor of Science in Electrical and Computer Engineering **GPA: 3.56/4.00**

SKILLS:

- **Software:** C, C++, Python, SQL, Verilog, MATLAB, Java, I2C/SPI/UART, RISC-V, Linux, Unix, Data Visualization, LTSpice, Cadence
- **Hardware:** PCB Design, FPGAs, Microcontrollers, Linear Circuit Design, Prototyping, Arduino, Raspberry Pi, MEMS Sensor Design
- **Misc. Technical:** Computer Architecture, Fusion360, OnShape CAD, ArcGIS, COMSOL Multiphysics, Jira, Confluence, Git, Github

RELEVANT COURSES AND PROJECTS:

- **IoT Traffic Sensor:** Developed an edge AI traffic monitoring system using Jetson Nano. Developed a YOLOv8 computer vision pipeline for vehicle classification, transmitted metadata over LoRaWAN, and optimized privacy by processing video locally.
- **MEMS Sensor Design and Simulation**: Designed, simulated, and optimized an inertial MEMS 3-axis gyroscope sensor to meet project quality requirements. Used COMSOL Multiphysics and L-Edit to improve sensor sensitivity and accuracy.
- **VLSI:** Used Cadence Virtuoso and Verilog to design, build, extract, and test a 512 by 512 DRAM array from transistor-level logic.
- **Courses**: Computer Architecture, Internet of Things, Microelectronics, VLSI, Embedded Systems, Digital Logic, Data Science, GPS Theory and Design, Microelectromechanical Systems (MEMS), Nanofabrication, Electromagnetic Fields and Waves.

EXPERIENCE:

Ecolectro: Electrical Engineering Intern **May 2025 – December 2025**
- Designed and implemented electrochemical cell test systems including a binary gas analyzer and flow mass sensor, Raspberry Pi-based monitoring to measure gas composition, voltage, current, and flow rates with real-time data logging.
- Automated test stand sensor data collection, wrote Python and C++ code for IV curve generation and automated calibration.
- Created comprehensive documentation to support long-term project continuity, presented final results to the CEO and CTO.

Arm: Hardware Engineering Intern **May 2024 – August 2024**
- Created dozens of Python scripts to automate FOM data analysis for transistor-level PDK computer hardware simulations.
- Analyzed PPA using Pandas and MatPlotLib to determine power consumption trends for 5, 3, and 2nm-transistor SOCs.
- Abstracted analytic capabilities of simulation data and significantly improved PPA analysis efficiency metrics.

Cornell University ECE 2400: Teaching Assistant **January 2024 – May 2024**
- Supported students in **ECE 2400: Computer Systems Programming**, a Cornell course focused on C/C++, systems-level programming, data structures, algorithms, and the connection between low-level hardware and user-facing software.
- Led office hours and helped students debug assignments in C/C++, memory management, Git, testing, and algorithms.
- Graded assignments, reviewed code for correctness and style, and explained core computer systems concepts to students.

CU GeoData Project Team Tech Team: Lead and Researcher **September 2022 – May 2026**
- Led a team of eight students to build a custom IOT sensor network to conduct earth and atmospheric science research.
- Built a custom Raspberry Pi NDVI-capable camera and drone system to collect vegetation health data around Ithaca, NY.
- Designed and built an Arduino-based temperature and contaminant probe to monitor and study harmful algae blooms.

American Museum of Natural History: Research Intern **August 2020 – June 2021**
- Worked with an AMNH research laboratory studying astrobiology and climate systems on Earth and exoplanets.
- Developed GIS projections using Panoply to predict and analyze future climate trends using large-scale climate datasets.
- Presented research at NYC Science Research Mentorship Consortium to dozens of professors, researchers, and students.

New York University: Research Intern **June 2020 – August 2020**
- Worked with the Music and Audio Research Lab to analyze, report on, and predict noise pollution trends in New York City.
- Implemented a comprehensive SQL database with 1000+ tagged audio samples to train machine learning predictive models.
- Coded in Python with Pandas, created and data visualization and prediction tools to present on noise pollution findings.

ADDITIONAL INFORMATION

- Published co-author at the American Geophysical Union Conference 2025
- Cornell Traditions Fellowship — Fall 2022 - Spring 2026
- Graduated cum laude